

院教学指导委员会主任 (院长)	学院分管教学 (副院长)	审核人 (专业负责人)	执笔人

中国武汉科技大学与英国伯明翰城市大学合作举办 网络工程专业本科教育项目培养方案

**Academic Plan and Curriculum for Undergraduate Network Engineering Program
Jointly Offered by Wuhan University of Science and Technology,
PRC and Birmingham City University, UK**

一、专业介绍

经教育部批准，武汉科技大学与英国伯明翰城市大学（Birmingham City University）合作举办网络工程专业本科项目，项目批准书编号为MOE42UK2A20141630N，办学层次和类别为本科学历教育，纳入国家普通高等教育招生计划。

本项目由中英双方共同制定人才培养方案和教学计划，并共同选派优秀教师任教，主要课程采用英语或双语教学，其中英方承担约1/3核心课程的教学。修业年限为4年，学生可在武汉科技大学完成全部学业，修完规定课程学分并考核合格，可获得武汉科技大学本科毕业证书和学士学位和伯明翰城市大学学士学位证书。

I. Professional Introduction

Upon approval of the Ministry of Education, PRC (Program Approval No. MOE42UK2A20141630N) and pursuant to the Co-operative Education Agreement signed between Wuhan University of Science and Technology, P. R. China and Birmingham City University, UK, the two parties have agreed to jointly offer the Bachelor of Engineering Program in Network Engineering beginning from 2015 for 10 years. The entry requirements of the Program are: (a) successful completion of the three-year senior secondary education in China, and (b) meeting the admissions requirements of WUST in the national College Entrance Examination.

The Program operates based on an Academic Plan and Curriculum jointly designed by both universities, which also provide qualified and experienced faculty to deliver teaching in the joint program up to due share (WUST 2/3 vs. BCU1/3 for core courses). The medium of instruction of the courses will be English or English complemented by Chinese on the part of the WUST faculty. The duration of study is 4 years. Students can complete all their studies in Wuhan University of Science and Technology, and after completing the required course credits and passing the assessment, they will be awarded the undergraduate certificate and bachelor's degree of Wuhan University of Science and Technology and the bachelor's degree of Birmingham City University.

二、培养目标

本专业围绕学校“钢铁品质、社会英才”的人才培养总目标，培养具有良好的人文素养、国际视野和创新精神，适应计算机网络技术发展，具备扎实的网络工程专业理论知识和专业技能，具有较强团队协作能力和终身学习能力，使之成为德智体美劳全面发展的社会主义事业合格建设者和可靠接班人。能够在互联网产业、钢铁冶金、大数据分析、金融贸易、物联网等行业的网络相关领域部门中，从事网络工程规划与设计、网络应用开发与实现、网络系统管理与维护、网络安全测评与防御等工作，担任企事业单位的技术或管理骨干的高素质应用型人才。

本专业毕业生工作5年左右预期达到的职业能力如下：

1. 具有良好的人文社科素养、社会责任感，遵守职业道德和规范，能够在计算机网络及相关领域的工程项目实践中，考虑对社会、健康、安全、法律、文化以及环境等因素的影响。
2. 能够综合运用数学、自然科学和专业基础知识，解决计算机网络及相关领域在网络工程规划与设计等工程实践中遇到的复杂工程问题。
3. 在互联网产业、钢铁冶金、大数据分析、金融贸易、物联网等行业的计算机网络相关领域部门中，能够胜任网络工程规划与设计、网络应用开发与实现、网络系统管理与维护、网络安全测评与防御等工作。
4. 具有良好的表达、沟通、协调和领导能力，能在多学科、多文化背景下组织和管理本领域的项目团队。
5. 具有终身学习能力，能够主动适应社会发展和职业发展的需求变化，通过自主学习不断更新专业知识，提升个人能力和技术水平。
6. 具有国际视野和良好的外语适应能力。

II. Training objectives

Focusing on the general goal of "steel quality, social talents", this major aim to cultivate students with good humanistic quality, international vision and innovative spirit, adapt to the development of computer network technology, have solid theoretical knowledge and professional skills of network engineering, and have strong teamwork ability and lifelong learning ability. Make it become a qualified builder and reliable successor of the socialist cause of all-round development of morality, intelligence, physique, beauty and labor. In the Internet industry, iron and steel metallurgy, big data analytics, financial and trade, the Internet of things industry such as network related departments, engaged in network engineering planning and design, application development and implementation, network system management and maintenance, network security assessment and defense work, as enterprises and institutions of the technical or management backbone of high-quality applied talents.

Graduates of this major are expected to achieve the following professional abilities in about 5 years of work:

1. Have good humanities and social science literacy, social responsibility, abide by professional ethics and norms, and be able to consider the impact on social, health, safety, legal, cultural and environmental factors in the practice of computer network and related engineering projects.
2. Be able to use mathematics, natural science and professional knowledge to solve complex engineering problems encountered in network engineering planning and design in computer network and related fields.
3. Be competent in network engineering planning and design, network application development and implementation, network system management and maintenance, network security evaluation and defense, etc. in computer network related departments in the Internet industry, iron and steel metallurgy, big data analysis, finance and trade, Internet of Things and other industries.

4. Good expression, communication, coordination and leadership skills, be able to organize and manage project teams in this field in a multi-disciplinary and multi-cultural context.

5. With the lifelong learning ability, I can actively adapt to the changes of social development and career development demands, constantly update my professional knowledge and improve my personal ability and technical level through independent learning.

6. Possesses an international perspective and excellent foreign language skills.

三、专业特色

本专业通过引进国际先进的教育方法以及资源，补充武汉科技大学现有相似专业，并向学生提供具有国际水准的教育；同时，希望为合作双方学生和员工提供教育和文化交流的机会，为两国的友好关系做出积极的贡献。

III. Professional characteristics

This program supplements the existing similar majors of Wuhan University of Science and Technology by introducing international advanced educational methods as well as resources, and provides students with education of international standards; at the same time, it hopes to provide students and employees of both sides of the cooperation with opportunities for educational and cultural exchanges, and to make a positive contribution to the friendly relations between the two countries.

四、毕业要求

1. 工程知识：能够将数学、自然科学、计算、工程基础和专业知用于解决复杂工程问题。
2. 问题分析：能够应用数学、自然科学和工程科学的基本原理，识别、表达并通过文献研究分析复杂工程问题，综合考虑可持续发展的要求，以获得有效结论。
3. 设计/开发解决方案：能够针对复杂工程问题设计和开发解决方案，设计满足特定需求的系统、单元（部件）或工艺流程，体现创新性，并从健康、安全与环境、全生命周期成本与净零碳要求、法律与伦理、社会与文化等角度考虑可行性。
4. 研究：能够基于科学原理并采用科学方法对复杂工程问题进行研究，包括设计实验、分析与解释数据、并通过信息综合得到合理有效的结论。
5. 使用现代工具：能够针对复杂工程问题，开发、选择与使用恰当的技术、资源、现代工程工具和信息技术工具，包括对复杂工程问题的预测与模拟，并能够理解其局限性。
6. 工程与可持续发展：在解决复杂工程问题时，能够基于工程相关背景知识，分析和评价工程实践对健康、安全、环境、法律以及经济和社会可持续发展的影响，并理解应承担的责任。
7. 工程伦理和职业规范：有工程报国、为民造福的意识，具有人文社会科学素养和社会责任感，能够理解和践行工程伦理，在工程实践中遵守工程职业道德、规范和相关法律，履行责任。
8. 个人与团队：能够在多样化、多学科背景下的团队中承担个体、团队成员以及负责人的角色。
9. 沟通：能够就复杂工程问题与业界同行及社会公众进行有效沟通和交流，包括撰写报告和设计文稿、陈述发言、清晰表达或回应指令；能够在跨文化背景下进行沟通和交流，理解、尊重语言和文化差异。
10. 项目管理：理解并掌握与工程项目相关的管理原理与经济决策方法，并能够在多学科环境中应用。
11. 终身学习：具有自主学习和终身学习的意识，有不断学习和适应发展的能力。

IV. Requirements

1. Engineering Knowledge: Ability to apply mathematical, natural science, computational, engineering

fundamentals, and specialized knowledge to solve complex engineering problems.

2. Problem Analysis: Ability to apply the fundamental principles of mathematics, natural science, and engineering science to identify, articulate, and analyze complex engineering problems through literature research, considering sustainability requirements comprehensively to arrive at effective conclusions.

3. Design/Development of Solutions: Ability to design and develop solutions for complex engineering problems, including systems, components, or processes that meet specific requirements, demonstrating innovation, and considering feasibility from perspectives such as health, safety, and environment; lifecycle cost and net-zero carbon requirements; legal and ethical considerations; and social and cultural factors.

4. Research: Ability to conduct research on complex engineering problems based on scientific principles and using scientific methods, including designing experiments, analyzing and interpreting data, and synthesizing information to reach reasonable and effective conclusions.

5. Use of Modern Tools: Ability to develop, select, and use appropriate technologies, resources, modern engineering tools, and information technology tools for complex engineering problems, including prediction and simulation of complex engineering problems, and understanding their limitations.

6. Engineering and Sustainable Development: When addressing complex engineering problems, be able to analyze and evaluate the impact of engineering practices on health, safety, the environment, legal compliance, and economic and social sustainability based on engineering-related background knowledge, and understand the responsibilities that must be assumed.

7. Engineering ethics and professional standards: Have a sense of engineering for the benefit of the country and the people, possess humanities and social science literacy and a sense of social responsibility, understand and practice engineering ethics, adhere to engineering professional ethics, standards, and relevant laws in engineering practice, and fulfill responsibilities.

8. Individual and team: Be able to assume the roles of an individual, team member, and leader in diverse, multidisciplinary teams.

9. Communication: Be able to effectively communicate and interact with industry peers and the general public on complex engineering issues, including writing reports and design documents, making presentations, clearly expressing or responding to instructions; be able to communicate and interact in cross-cultural contexts, understanding and respecting language and cultural differences.

10. Project Management: Understand and master management principles and economic decision-making methods related to engineering projects, and be able to apply them in multidisciplinary environments.

11. Lifelong Learning: Possess the awareness of self-directed and lifelong learning, and the ability to continuously learn and adapt to development.

附：培养目标实现矩阵

	培养目标 1	培养目标 2	培养目标 3	培养目标 4	培养目标 5	培养目标 6
毕业要求 1		√				
毕业要求 2		√	√			
毕业要求 3		√	√			
毕业要求 4			√			

	培养目标 1	培养目标 2	培养目标 3	培养目标 4	培养目标 5	培养目标 6
毕业要求 5			√		√	
毕业要求 6	√					√
毕业要求 7	√					√
毕业要求 8	√					
毕业要求 9				√		√
毕业要求 10				√	√	
毕业要求 11				√		√

五、专业主干课程

计算机组成原理、数据结构、操作系统、计算机网络、网络交换技术、路由技术、高级网络设计、网络安全。

V. Core courses

Principles of Computer Composition, Data Structure, Operating System, Computer Network, Network Switched Technology, Routing Technology, Advanced Network Design, Network Simulation and Modeling, Cyber Security.

六、基本学制：四年

VI. Recommended length of the program: 4 years of full-time study

七、授予学位：工学学士

修业年限为4年，学生可在武汉科技大学完成全部学业，修完规定课程学分并考核合格，可获得武汉科技大学本科毕业证书和工学学士学位和伯明翰城市大学学士学位证书。

VII. Degree: Bachelor of Engineering

The duration of study is 4 years. Students can complete all their studies in Wuhan University of Science and Technology, and after completing the required course credits and passing the assessment, they will be awarded the undergraduate certificate and bachelor's degree of Wuhan University of Science and Technology and the bachelor's degree of Birmingham City University.

八、毕业学分要求：180学分

课程类型		学分要求	课程类型	学分要求	
1、公共课程平台		53	3、专业课程模块	64.5	
公共基础课程		19	专业必修课程	60.5	
通识教育课程	必修	28	专业选修课程	4	
	选修	6	4、实践教学模块	24	
2、学科基础平台		32.5	专业实践课程	必修	24
专业学科基础课程	必修	30.5		选修	0
	选修	2	5、素质拓展模块		6

*通识教育选修课6学分包括：理工类、医学类、人文社科类、经济管理类中选择2学分（学生在本专业所属类别外的3个类中选择2个学分）；思想政治及新时代素质教育类选择2学分（“四史”至少选修1个学分）；美育教育类选择2学分（美学和艺术史论类、艺术鉴赏和评论类课程至少选修1个学分）。

VIII. Credits required for graduation : 180 credits

Type of courses		Academic credits	Type of courses	Academic credits
1.Common Courses		53	3. Specialized Courses	64.5
Common Basic Courses		19	Required Courses	60.5
General Education Courses	Required Courses	28	Elective Courses	4
	Elective Courses	6	4.Practicum and Internship Courses	24
2.General Disciplinary Courses		32.5	Disciplinary Required Courses	24
Disciplinary Basic Courses	Required Courses	30.5	Practical Courses	0
	Elective Courses	2	5.Quality Development Courses	6

九、学分比例

IX. Ratio of Credits

1、必修选修学分比例

The proportion of compulsory elective credits

类别 Type	学分 Credit	占总学分比例 Ratio
必修 Required	168	93.3%
选修 elective	12	6.7%

2、实践教学环节学分比例

The Proportion of credits in practice teaching

类别 Type	学分 Credit	占总学分比例 Radio
实践教学环节 Practical Teaching Sessions	实验教学学分 Experiment	16.8
	实践教学模块 Practice	24.0
	素质拓展模块 Quality development program	6
		26.0%

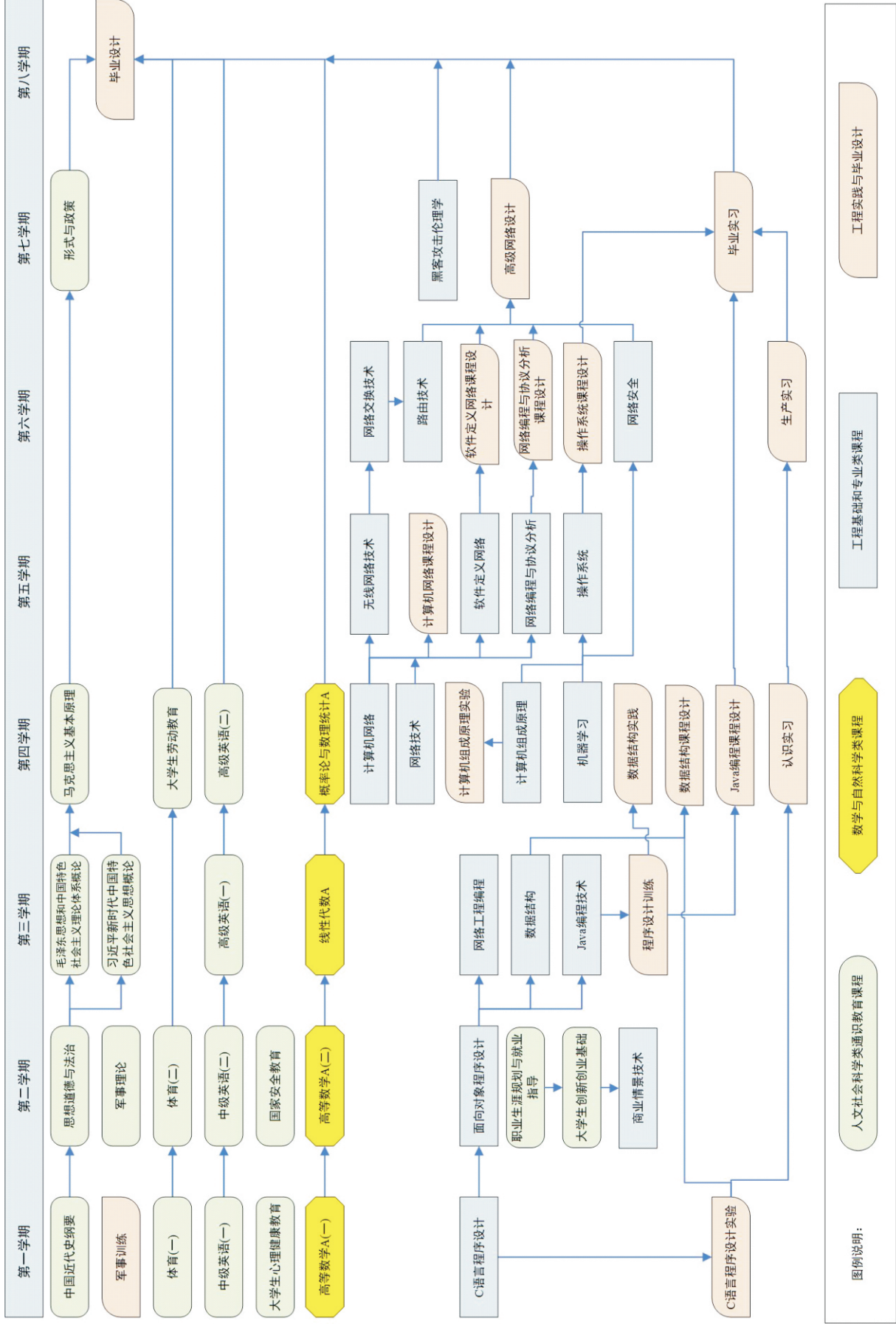
十、毕业要求实现矩阵

X. Graduation Realization Matrix

课程名称	网络工程专业毕业要求										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
中国近现代史纲要							√				
思想道德与法治						√	√				
毛泽东思想和中国特色社会主义理论体系概论						√	√				
习近平新时代中国特色社会主义思想概论						√	√				
马克思主义基本原理						√	√				√
形势与政策						√					
国家安全教育						√			√		
中级英语									√		
高级英语									√		
体育								√			
军事理论							√	√			
军事训练							√	√			
大学生心理健康教育						√					
职业生涯规划与就业指导						√	√				√
大学生创新创业基础							√	√			√
C 语言程序设计	√	√	√								
面向对象程序设计	√	√	√								
高等数学 A	√	√		√							
线性代数	√	√		√							
概率论与数理统计 A	√	√		√							
商业情景技术		√									
网络工程编程		√									
数据结构	√	√	√								
JAVA 程序设计		√									
计算机组成原理	√	√		√							
网络技术		√	√								
计算机网络			√								
机器学习			√		√					√	

课程名称	网络工程专业毕业要求										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
操作系统	√	√		√							
无线网络技术			√		√						
网络编程与协议分析			√						√		
软件定义网络		√	√								
网络交换技术			√								
路由技术			√								
网络安全			√		√						
高级网络设计			√						√		
黑客攻击伦理学			√			√					
C 语言程序设计实验				√	√				√		
程序设计训练				√	√				√		
数据结构实践		√	√		√				√		
认识实习				√		√			√		
数据结构课程设计		√	√		√				√		
计算机组成原理实验				√	√				√		
计算机网络课程设计				√	√				√		
操作系统课程设计				√	√				√		
网络编程与协议分析课程设计				√	√				√		
软件定义网络课程设计				√	√				√		
生产实习				√	√	√	√	√	√	√	
毕业实习				√	√	√	√	√	√	√	
毕业设计(论文)			√	√	√	√	√		√		√
创新创业实践								√		√	√
第二课堂								√	√		
大学生劳动教育						√		√			

十一、课程修读进程表
XI. Course review Process Map
(四年制本科 2025级 网络工程专业“中外合作办学”)



十二、教学环节设置及学分分布表 (四年制本科 2025级 网络工程专业“中外合作办学”)

XII. Offered Course and Distribution of Academic Credits

课 程 类 型	课程 性质	课 程 编 码	课 程 名 称	学 分	合 计	课 内 学 时			实 践 学 时	学 期	是否 辅修/ 双学位 课程	先 修 课 程/备 注
						讲 课	实 验	上 机				
平台	公共基础课程	必修	5105017 思想道德与法治 Ideology and morality and rule of law	3	48	42	0	0	6	2		
			5103001 中国近现代史纲要 An Outline of Modern and Contemporary History of China	3	48	42	0	0	6	1		
			5102001 马克思主义基本原理 Fundamentals of Marxism	3	48	44	0	0	4	4		
			5101005 毛泽东思想和中国特色社会主义理论体系概论 Theoretical system of socialism with Chinese characteristics	3	48	32	0	0	16	3		
			5101006 习近平新时代中国特色社会主义思想概论 An Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3	48	32	0	0	16	3		
			1501882 体育(一) Physical Education (I)	1	26	26	0	0	0	1		
			1501883 体育(二) Physical Education (II)	1	34	34	0	0	0	2		
			5106001 形势与政策 World Affairs and State Policy	2	64	64	0	0	0	1-8		分散进行
			1501024 体质健康测试 Physical Health Test	0	16	0	0	0	16	1-8		分散进行
	公共课程平台课程	必修	5107001 国家安全教育 National Security Education	1	16	8	0	0	8	2		
			2501016 大学生劳动教育 College student labor education	1	32	2	0	0	30	1-4		分散进行
			2104060 中级英语(一) Intermediate English (I)	6	96	96	0	0	0	1		
			2104061 中级英语(二) Intermediate English (II)	6	96	96	0	0	0	2		
			2104062 高级英语(一) Advanced English (I)	4	64	64	0	0	0	3		
			2104063 高级英语(二) Advanced English (II)	4	64	64	0	0	0	4		
			2502006 大学生心理健康教育 Mental Health Education	2	32	24	0	0	8	1		

教学环节设置及学分分布表

(四年制本科 2025级 网络工程专业“中外合作办学”)

Offered Course and Distribution of Academic Credits

课 程 类 型	课程 性质	课 程 编 码	课 程 名 称	学 分	合 计	课 内 学 时			实 践 学 时	学 期	是否 辅修/ 双学位 课程	先 修 课 程/备 注
						讲 课	实 验	上 机				
平台	公共课程平台课程	必修	2503001 职业生涯规划与就业指导 Career Planning and Employment Guidance	1	16	16	0	0	0	2		
			2504005 军事理论 Military Theory	2	36	36	0	0	0	2		
			8001003 大学生创新创业基础 Foundation of Innovation and Entrepreneurship for college students	1	16	16	0	0	0	2		
		选修	医学类1学分 Medicine 1 Academic Credit									
			人文社科类1学分 Humanity and Social Science 1 Academic Credit									
			经济管理类1学分 Economic and Management 1 Academic Credit									
			思想政治及新时代素质教育类2学分 Ideological and Political Education 2 Academic Credit									
			美育教育类2学分 Art Education 2 Academic Credit									
	专业学科基础平台课程	必修	0702601 高等数学A(一) Advanced Mathematics A (I)	5	80	80	0	0	0	1		
			1303068 C语言程序设计 C Programming	3	48	48	0	0	0	1		
			1303098 面向对象程序设计 Object-Oriented Programming	2.5	40	32	0	8	0	2		
			0702602 高等数学A(二) Advanced Mathematics A (II)	6.5	104	104	0	0	0	2		
			1304077 商业情景技术★ Context of business Technology★	4.5	72	62	0	10	0	2		
			1304100 网络工程编程★ Programming for Network Engineering★	3	48	38	0	10	0	3		
			0702103 线性代数A Linear Algebra A	3	48	48	0	0	0	3		
			0702303 概率论与数理统计A Probability Theory and Mathematical Statistics A	3	48	48	0	0	0	4		

教学环节设置及学分分布表

(四年制本科 2025级 网络工程专业“中外合作办学”)

Offered Course and Distribution of Academic Credits

课 类	程 型	课程 性质	课 程 编 码	课 程 名 称	学 分	合 计	课 内 学 时			实 践 学 时	学 期	是否 辅修/ 双 学 位 课 程	先 修 课 程/备 注
							讲 课	实 验	上 机				
平台	学科 基础 平台	专业 学科 基础 课程	选修	1304046	网络科学与工程导论 Introduction to Network Science and Engineering	2	32	32	0	0	0	2	
				1303096	离散数学 Discrete Mathematics	3	48	48	0	0	0	2	
模块			必修	1302133	数据结构 Data Structure	3.5	56	56	0	0	0	3	是 C语言程序 设计
				1304096	JAVA 程序设计★ JAVA Programming★	5	80	80	0	0	0	3	
				1303023	计算机组成原理 Principles of Computer Organization	3.5	56	56	0	0	0	4	
				1304101	网络技术 ★ Networking Technologies★	4	64	50	14	0	0	4	
				1304102	计算机网络★ Computer Network★	5	80	66	14	0	0	4	
				1304103	机器学习★ Machine Learning★	4	64	50	0	14	0	4	AI+
				1302134	操作系统 Operating System	2.5	40	40	0	0	0	5	
				1304104	无线网络技术★ Wireless Networking Technology★	3	48	40	0	8	0	5	
				1304035	网络编程与协议分析★ Network Programming and Protocol Analysis★	4	64	50	0	14	0	5	
				1304097	软件定义网络 ★ Software Design Network★	4	64	50	0	14	0	5	
				1304084	网络交换技术★ Network Switched Technology★	5	80	66	14	0	0	6	
				1304105	路由技术 ★ Routing Technology★	5	80	66	14	0	0	6	
				1304106	网络安全 ★ Cyber Security★	5	80	66	14	0	0	6	AI+
				1304110	高级网络设计 ★ Advanced Network Design★	3	48	40	18	0	0	7	
				1304109	黑客攻击伦理学 ★ Ethical Hacking★	4	64	50	14	0	0	7	7

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						讲 课	实 验	上 机				
模块	专业 课程 模块	专业 选修 课程	1304090 知识图谱技术与应用 Knowledge graph technique and application	2	32	32	0	0	0	5		
			1304091 物联网技术 Internet of Things	2	32	24	8	0	0	6		
			1304070 大数据技术原理与应用 Big Data Technology and Applications	2	32	26	0	6	0	6		
	实践教学 模块	专业 实践 课程 必修	1303008 C语言程序设计实验 Experiments in C Programming	1	24	0	0	24	0	1		
			2504006 军事训练 Military Training	2	112	0	0	0	112	1		
			1303015 程序设计训练★ Advanced Training for Programming Design★	1	1 (周)	0	0	0	1 (周)	3		
			1302006 数据结构实践★ Data Structure Practice★	1	1 (周)	0	0	0	1 (周)	4		
			1304073 认识实习 Cognition Practice	1	1 (周)	0	0	0	1 (周)	4		
			1302034 数据结构课程设计 Course Project in Data Structure	1	1 (周)	0	0	0	1 (周)	4		
			1303009 计算机组成原理实验 Experiments in Principles of Computer Organization	1	24	0	24	0	0	4		
			1304022 计算机网络课程设计★ Course Project in Ethernet ★	1	1 (周)	0	0	0	1 (周)	5		
			1302122 操作系统课程设计 Course Project in Operating System	1	1 (周)	0	0	0	1 (周)	6		
			1304036 网络编程与协议分析课程设计★ Course Project in Network Programming and Protocol Analysis★	1	1 (周)	0	0	0	1 (周)	6		
			1304099 软件定义网络课程设计★ Course Project in Software Design Network★	1	1 (周)	0	0	0	1 (周)	6		
			1304074 生产实习 Production Practice	2	2 (周)	0	0	0	2 (周)	6		
			1304033 毕业实习 Pre-graduation Internship	2	2 (周)	0	0	0	2 (周)	7		

教学环节设置及学分分布表

(四年制本科 2025级 网络工程专业“中外合作办学”)

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课 程 类 型	课程性质	课 程 编 码	课 程 名 称	学 分	合 计	课 内 学 时			实 践 学 时	学 期	是否 辅修/ 双学位 课程	先 修 课 程/备 注
						讲 课	实 验	上 机				
模块	实践教学模块	必修	1305028 毕业设计(论文) Graduation Project	8	14 (周)	0	0	0	14 (周)	8		
	创新创业教育	必修	创新创业实践3学分 Innovation Practices 3 Academic Credits									
	素质拓展模块	必修	第二课堂3学分 Second Classroom 3 Academic Credits									

*创新创业教育和第二课堂主要安排在小学期进行